SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

for

FREELANCE MARKETPLACE

Web Application Platform

IEEE 830-1998 Compliant

Version 1.0

January 2025

Academic Project Documentation

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Document Revision History

Version	Date	Author	Description
1.0	January 2025	Development Team	Initial SRS document creation

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the Freelance Marketplace web application. It details the functional and non-functional requirements, system constraints, and interface specifications necessary for the development team to design and implement the platform.

The intended audience includes software developers, project managers, quality assurance teams, system architects, and academic evaluators who will use this document as the primary reference for understanding system requirements and validating the final implementation.

1.2 Scope

The Freelance Marketplace is a web-based platform that facilitates connections between freelance professionals and clients seeking specialized services. The system will provide an end-to-end solution for the freelance work lifecycle, from project posting to payment completion.

Key objectives of the system include:

- Enable clients to post projects and hire qualified freelancers
- Allow freelancers to showcase skills and bid on available projects
- Provide secure payment processing with escrow protection
- Facilitate real-time communication between parties
- Implement a transparent review and rating system
- Offer administrative tools for platform management

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token

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CRUD	Create, Read, Update, Delete operations
UI/UX	User Interface / User Experience
ORM	Object-Relational Mapping
Escrow	Third-party holding of funds until conditions are met
Milestone	A significant checkpoint in project progress with associated payment
Proposal/Bid	A freelancer's offer to complete a project at a specified price
WebSocket	Protocol for real-time bidirectional communication

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- IEEE Std 1016-2009: IEEE Standard for Information Technology - Systems Design
- OWASP Security Guidelines for Web Applications
- PCI DSS v4.0: Payment Card Industry Data Security Standard
- GDPR: General Data Protection Regulation compliance requirements

1.5 Overview

This document is organized into seven main sections. Section 2 provides an overall description of the product including its context, functions, and constraints. Section 3 details external interface requirements. Section 4 describes specific system features with detailed functional requirements. Section 5 covers non-functional requirements. Section 6 addresses other requirements, and Section 7 contains appendices with supplementary information.

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2. Overall Description

2.1 Product Perspective

The Freelance Marketplace is a self-contained web application that operates independently but integrates with external services for enhanced functionality. The system follows a client-server architecture with a clear separation between the presentation layer, business logic, and data persistence.

2.1.1 System Architecture

The application employs a three-tier architecture comprising the Presentation Tier (web browser client), Application Tier (web server with business logic), and Data Tier (database server). This architecture supports modularity, maintainability, and scalability.

2.1.2 External System Interfaces

- **Payment Gateway:** Integration with Stripe for secure payment processing
- **Email Service:** SMTP integration for transactional emails and notifications
- **Cloud Storage:** AWS S3 or Cloudinary for file and media storage
- **OAuth Providers:** Google and GitHub for social authentication

2.2 Product Functions

The following diagram illustrates the major functional areas of the system:

Functional Area	Description
User Management	Registration, authentication, profile management, role-based access
Project Management	Create, edit, search, filter, and manage project listings
Bidding System	Submit, modify, withdraw proposals; accept/reject bids
Contract Management	Contract creation, milestone tracking, deliverable submission
Messaging	Real-time chat, notifications, file sharing
Payments	Escrow, milestone payments, withdrawals, invoicing
Reviews & Ratings	Post-project feedback, rating aggregation, reputation system

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Administration	User management, dispute resolution, analytics, configuration

2.3 User Classes and Characteristics

2.3.1 Freelancer

Attribute	Description
Description	Professional offering services on the platform
Technical Expertise	Moderate; comfortable with web applications
Frequency of Use	Daily to weekly
Key Functions	Profile management, project search, bidding, deliverable submission, payment withdrawal

2.3.2 Client

Attribute	Description
Description	Individual or business seeking freelance services
Technical Expertise	Low to moderate; basic computer literacy required
Frequency of Use	Occasional to frequent depending on project needs
Key Functions	Project posting, proposal review, hiring, milestone approval, payment release

2.3.3 Administrator

Attribute	Description
Description	Platform staff responsible for system management
Technical Expertise	High; trained on administrative functions
Frequency of Use	Daily
Key Functions	User management, dispute resolution, content moderation, analytics, system configuration

2.4 Operating Environment

2.4.1 Client-Side Requirements

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- **Web Browsers:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **Screen Resolution:** Minimum 320px width (mobile responsive)
- **JavaScript:** Must be enabled
- **Internet Connection:** Broadband recommended for real-time features

2.4.2 Server-Side Requirements

- **Operating System:** Linux (Ubuntu 22.04 LTS recommended)
- **Database:** MySQL 8.0+
- **Cache:** Redis 7.0+
- **Memory:** Minimum 4GB RAM, 8GB recommended
- **Storage:** SSD with minimum 50GB available

2.5 Design and Implementation Constraints

- **Regulatory:** Must comply with GDPR for data protection and PCI DSS for payment processing
- **Technology:** Must use specified technology stack (React/Node.js or Django with MySQL)
- **Accessibility:** Must meet WCAG 2.1 Level AA compliance
- **Security:** Must implement OWASP Top 10 security controls
- **API Design:** Must follow RESTful conventions with JSON responses
- **Timeline:** Development constrained to 10-week academic semester

2.6 Assumptions and Dependencies

Assumptions

- Users have access to stable internet connectivity
- Users have basic computer literacy and web browsing experience
- Third-party services (Stripe, email, cloud storage) maintain uptime SLAs
- Users will provide valid email addresses for account verification

Dependencies

- Stripe API for payment processing functionality
- SMTP service for email delivery
- Cloud storage provider for file uploads
- OAuth providers (Google, GitHub) for social login
- SSL/TLS certificates for secure communication

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3. External Interface Requirements

3.1 User Interfaces

3.1.1 General UI Requirements

- Responsive design supporting desktop, tablet, and mobile devices
- Consistent navigation structure across all pages
- Clear visual hierarchy with intuitive information architecture
- Loading indicators for asynchronous operations
- Informative error messages with recovery guidance
- Confirmation dialogs for destructive actions

3.1.2 Key Screens

Screen	Description
Landing Page	Platform introduction, featured projects, CTA for registration
Registration/Login	User authentication forms with social login options
Dashboard	Personalized overview of activities, notifications, quick actions
Project Listing	Searchable, filterable list of available projects
Project Detail	Full project information, bid submission form
Profile Page	User profile with portfolio, reviews, work history
Messages	Real-time chat interface with conversation list
Contract View	Active contract details, milestones, deliverables
Payment Center	Transaction history, withdrawal requests, invoices

3.2 Hardware Interfaces

The system is a web application and does not directly interface with hardware. Users access the system through standard web browsers on their devices. The server infrastructure may utilize network interfaces, storage systems, and load balancers as managed by the hosting environment.

3.3 Software Interfaces

External System	Interface Type	Purpose
Stripe API	REST API	Payment processing, escrow management, payouts

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MySQL Database	ORM/SQL	Persistent data storage and retrieval
Redis	TCP Protocol	Session management, caching, pub/sub for real-time
AWS S3/Cloudinary	REST API	File upload, storage, and CDN delivery
SMTP Server	SMTP Protocol	Transactional email delivery
Google OAuth	OAuth 2.0	Social authentication
GitHub OAuth	OAuth 2.0	Social authentication for developers

3.4 Communication Interfaces

- **HTTP/HTTPS:** Primary protocol for client-server communication (TLS 1.3)
- **WebSocket:** Bidirectional communication for real-time messaging and notifications
- **JSON:** Data interchange format for API requests and responses
- **SMTP/TLS:** Secure email transmission

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4. System Features

4.1 User Authentication and Management

Priority: High

4.1.1 Description

This feature provides secure user registration, authentication, and profile management capabilities. It supports multiple authentication methods and implements role-based access control.

4.1.2 Functional Requirements

ID	Requirement	Priority
SR-AUTH-001	System shall allow users to register with email and password	High
SR-AUTH-002	System shall send email verification upon registration	High
SR-AUTH-003	System shall support OAuth login via Google and GitHub	Medium
SR-AUTH-004	System shall implement JWT-based session management	High
SR-AUTH-005	System shall provide password reset via email link	High
SR-AUTH-006	System shall enforce password complexity requirements	High
SR-AUTH-007	System shall support user role selection (Freelancer/Client)	High
SR-AUTH-008	System shall allow users to update profile information	High
SR-AUTH-009	System shall support profile picture upload	Medium
SR-AUTH-010	System shall implement account deactivation	Low

4.2 Project Management

Priority: High

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4.2.1 Description

This feature enables clients to create, manage, and track projects while allowing freelancers to discover and apply for relevant opportunities.

4.2.2 Functional Requirements

ID	Requirement	Priority
SR-PROJ-001	System shall allow clients to create projects with title, description, category	High
SR-PROJ-002	System shall support budget specification (fixed/hourly/range)	High
SR-PROJ-003	System shall allow file attachments to project descriptions	Medium
SR-PROJ-004	System shall support skill tagging for projects	High
SR-PROJ-005	System shall provide project search with filters	High
SR-PROJ-006	System shall implement project status workflow	High
SR-PROJ-007	System shall allow project editing before bids are accepted	Medium
SR-PROJ-008	System shall support project deadline specification	High
SR-PROJ-009	System shall display project view count	Low
SR-PROJ-010	System shall provide pagination for project listings	Medium

4.3 Bidding System

Priority: High

4.3.1 Description

This feature allows freelancers to submit proposals on projects and enables clients to review, compare, and accept bids.

4.3.2 Functional Requirements

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ID	Requirement	Priority
SR-BID-001	System shall allow freelancers to submit proposals with cover letter	High
SR-BID-002	System shall require bid amount and estimated timeline	High
SR-BID-003	System shall allow bid modification before client review	Medium
SR-BID-004	System shall support bid withdrawal	Medium
SR-BID-005	System shall notify freelancers when bid is viewed	Low
SR-BID-006	System shall allow clients to accept or reject bids	High
SR-BID-007	System shall create contract upon bid acceptance	High
SR-BID-008	System shall limit one active bid per freelancer per project	High

4.4 Messaging System

Priority: High

4.4.1 Functional Requirements

ID	Requirement	Priority
SR-MSG-001	System shall provide real-time messaging between users	High
SR-MSG-002	System shall support file attachments in messages	Medium
SR-MSG-003	System shall display read receipts for messages	Low
SR-MSG-004	System shall show typing indicators	Low
SR-MSG-005	System shall provide message search functionality	Medium
SR-MSG-006	System shall send email notifications for offline messages	Medium

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SR-MSG-007	System shall display online/offline status indicators	Low

4.5 Payment System

Priority: High

4.5.1 Functional Requirements

ID	Requirement	Priority
SR-PAY-001	System shall implement escrow-based payment holding	High
SR-PAY-002	System shall support milestone-based payments	High
SR-PAY-003	System shall integrate with Stripe for payment processing	High
SR-PAY-004	System shall release funds upon milestone approval	High
SR-PAY-005	System shall allow freelancers to request withdrawals	High
SR-PAY-006	System shall generate invoices for transactions	Medium
SR-PAY-007	System shall maintain complete transaction history	High
SR-PAY-008	System shall calculate and deduct platform fees	High

4.6 Review and Rating System

Priority: Medium

4.6.1 Functional Requirements

ID	Requirement	Priority
SR-REV-001	System shall allow reviews only after project completion	High
SR-REV-002	System shall support 1-5 star rating scale	High

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SR-REV-003	System shall require written feedback with rating	Medium
SR-REV-004	System shall calculate and display average ratings	High
SR-REV-005	System shall allow both parties to review each other	High
SR-REV-006	System shall prevent review editing after submission	Medium

4.7 Admin Dashboard

Priority: Medium

4.7.1 Functional Requirements

ID	Requirement	Priority
SR-ADM-001	System shall provide user management (view, suspend, delete)	High
SR-ADM-002	System shall display platform analytics and metrics	Medium
SR-ADM-003	System shall provide dispute resolution interface	High
SR-ADM-004	System shall allow category and skill management	Medium
SR-ADM-005	System shall provide transaction monitoring	High
SR-ADM-006	System shall support content moderation	Medium
SR-ADM-007	System shall generate downloadable reports	Low

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5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement	Metric
NFR-PERF-001	Page load time for static content	< 2 seconds
NFR-PERF-002	API response time for standard requests	< 500ms
NFR-PERF-003	Real-time message delivery latency	< 100ms
NFR-PERF-004	Concurrent user support	> 500 users
NFR-PERF-005	Database query response time	< 100ms
NFR-PERF-006	File upload processing time (5MB)	< 5 seconds

5.2 Security Requirements

ID	Requirement
NFR-SEC-001	All data transmission must use TLS 1.3 encryption
NFR-SEC-002	Passwords must be hashed using bcrypt with minimum cost factor of 12
NFR-SEC-003	System must implement CSRF protection on all state-changing requests
NFR-SEC-004	All user input must be sanitized to prevent XSS attacks
NFR-SEC-005	Database queries must use parameterized statements to prevent SQL injection
NFR-SEC-006	API endpoints must implement rate limiting (100 requests/minute per user)
NFR-SEC-007	Session tokens must expire after 24 hours of inactivity

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NFR-SEC-008	Failed login attempts must be limited (5 attempts, 15-minute lockout)
NFR-SEC-009	Payment data must never be stored on application servers
NFR-SEC-010	Security headers must be implemented (CSP, X-Frame-Options, etc.)

5.3 Reliability Requirements

ID	Requirement	Target
NFR-REL-001	Mean Time Between Failures (MTBF)	> 720 hours
NFR-REL-002	Mean Time To Recovery (MTTR)	< 1 hour
NFR-REL-003	Data backup frequency	Every 6 hours
NFR-REL-004	Backup retention period	30 days
NFR-REL-005	Transaction success rate	> 99.9%

5.4 Availability Requirements

ID	Requirement	Target
NFR-AVL-001	System uptime	99.5%
NFR-AVL-002	Planned maintenance window	< 4 hours/month
NFR-AVL-003	Maintenance notification lead time	> 24 hours

5.5 Scalability Requirements

ID	Requirement

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NFR-SCL-001	System shall support horizontal scaling of application servers
NFR-SCL-002	Database shall support read replicas for load distribution
NFR-SCL-003	Static assets shall be served via CDN
NFR-SCL-004	System shall support up to 10,000 registered users
NFR-SCL-005	System shall handle 1,000 concurrent active sessions

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6. Other Requirements

6.1 Legal and Regulatory Requirements

- GDPR compliance for user data handling and privacy
- PCI DSS compliance for payment card data (handled via Stripe)
- Terms of Service and Privacy Policy must be presented during registration
- User consent must be obtained for data processing activities
- Right to data portability and deletion must be supported

6.2 Usability Requirements

- New users shall complete registration within 3 minutes
- Users shall accomplish primary tasks with no more than 5 clicks
- Error messages shall provide clear guidance for resolution
- Interface shall be usable without external documentation
- System shall support keyboard navigation for accessibility

6.3 Documentation Requirements

- User guide for freelancers and clients
- Administrator manual for platform management
- API documentation for third-party integrations
- Installation and deployment guide
- FAQ and troubleshooting guide

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7. Appendices

Appendix A: Use Case Summary

UC ID	Use Case Name	Primary Actor
UC-001	Register Account	Guest User
UC-002	Login to System	Registered User
UC-003	Create Project	Client
UC-004	Search Projects	Freelancer
UC-005	Submit Proposal	Freelancer
UC-006	Review Proposals	Client
UC-007	Accept Bid	Client
UC-008	Send Message	Registered User
UC-009	Submit Deliverable	Freelancer
UC-010	Approve Milestone	Client
UC-011	Request Withdrawal	Freelancer
UC-012	Submit Review	Registered User
UC-013	Resolve Dispute	Administrator

Appendix B: Data Dictionary

Entity	Attribute	Description
User	id	Unique identifier (Primary Key)
User	email	Unique email address for authentication
User	role	User type (freelancer, client, admin)
Project	status	Current state (draft, open, in_progress, completed)
Project	budget_type	Payment structure (fixed, hourly, milestone)
Proposal	status	Bid state (pending, accepted, rejected, withdrawn)
Milestone	status	Progress (pending, submitted, approved, paid)

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Transaction	type	Transaction category (escrow, release, withdrawal)

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